MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CROSS FIELD AMPLIFIER

1/ TYPE 8898

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Broadband, pulsed, integral magnet, liquid cooled, 2,905 to 3,085 MHz, 666 kw rated peak power output.

ABSOLUTE RATINGS: See note 27

Parameter:	epy	epx	ib	tp	Du	VSWR	Waveguide pressurization	T(bushing)	TA
Unit:	kv	kv	a	µs	---	---	psia	°C	°C
Maximum:	60	20	23	33	0.01	-1.3	55	200	-54
Minimum:	---	---	18	---	0.006	---	40	---	+65
						(Note 1)	(Note 2)		

ABSOLUTE RATINGS: See note 27

	Coolant inlet		Coolant pressure		Coolant flow rate			
Parameter:	T(water)	T(EG & W)	Window	Anode	Window	Anode	Alt	F
Unit:	°C	°C	psia	psia	gpm	gpm	ft	MHz
Maximum:	65	14	55	75	---	---	10,000	F3
Minimum:	---	---	---	---	0.4	3.0	---	F1
	(Note 3)				(Note 3)			

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Cooling:	Forced liquid
Mounting position:	Cathode axis vertical	Handling:	See note 5
Input connection:	See note 4	Weight:	130 lb (max)
RF connectors:	Mates with UG-54B/U type flange per MIL-F-3922/26	Magnet isolation:	12 inches (see note 5)

TEST CONDITIONS: See notes 6, 12, and 13

Parameter:	Ib	Du	tp	VSWR	pd	Waveguide pressurization	F
Unit:	mAdc	---	µs	---	kw	psia	MHz
Tolerance:	---	---	±1	---	Max	Min	---
Test condition 1:	200	0.010	32	1.1	48	45	F1,F2,F3
Test condition 2:	124 (Note 7)	0.0062	32 (Note 8)	1.1 (Note 1)	48 (Notes 9,10)	45 (Note 2)	F1,F2,F3 (Note 12)

GENERAL:

Qualifications - Required

1/ Replaces - Raytheon type QKS-1594
ITT Gilfillan part number 145110-1

FREQUENCY	
F	MHz
1	2,905
2	3,000
3	3,085

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METHOD	REQUIREMENT OR TEST	NOTES	TESTS	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	6,11, 12						
4003	Pressurization (rf window)	2	-	55 psia (max)	---	---	---	---
4306	Pulse voltage	8	1,2	F1, F3	epy	45	55	kv
4250	Power output	13,14 13,14	1 2	F1	Po Po	6,600 4,160	--- ---	W W
4279	Reverse directed power	13,19	1	VSWR = 1.3:1 (min)	Pr	---	1,110	W
---	Stability	13,15, 16	1,2		MP	---	40	Pulses
---	Starting stability	13,16, 20	1	ib(start) = 20 a	MP	---	40	Pulses
---	Stability (mode)	13,17	1	VSWR=1.3:1 (min); Ib = 110 mAdc at F1 and F3	---	---	---	---
---	Amplifier insertion loss	10	-	pd = 48 kw (see figure 2)	Li	---	3.0	dB
1261	Electrode voltage (getter)	-	-	$I_{(getter)}$ = 15 A; t = 300 (min)	E (getter)	1.0	5.0	V
	Quality conformance inspection, part 2							
4266	Direct-inter-electrode capacitance	-	-		C	---	50	pF
1155	Coolant-pressure drop versus coolant flow	3	-					
				Water				
	Anode	-	-	Flow rate = 3 gpm	ΔP	6	20	psi
	Window	-	-	Flow rate = 0.4 gpm	ΔP	0.5	6	psi
				Ethylene glycol and water (EG&W)				
	Anode	-	-	Flow rate = 4 gpm	ΔP	10	35	psi
	Window	-	-	Flow rate = 0.5 gpm	ΔP	0.5	7	psi
1105	Permanence of marking	-	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	TESTS	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
---	Life test	18, 21,22	1	Group D	t	1,000	---	hrs
---	Life-test end points:							
4250	Power output	13,14	1		Po	6,600	---	W
---	Stability	13,15, 16	1,2		MP	---	40	Pulses
	Periodic check tests							
1032	Vibration, mechanical	23,24	-	No voltages	---	---	---	---
1042	Shock, specified pulse	24,25	-	Test condition A	---	---	---	---
				Test condition J	---	---	---	---
---	Temperature, altitude	24,26	-	Temperature, operating	T	-54	+60	°C
				Temperature, non-operating	T	-65	+65	°C
				Altitude, operating	---	10,000	---	ft
				Altitude, non-operating	---	30,000	---	ft

NOTES:

1. The system match looking away from the input and output flanges shall have a VSWR less than 1.5:1 for the frequency range of 2,600 to 3,200 MHz. The VSWR shall be less than 1.3:1 for the operating range of F1 to F3.

2. The pressurization gas shall provide insulating properties which are the equivalent of oil-filtered and particle-filtered air dried to a dew point below 40°F. The oil trap may be activated alumina, the filter may be 20 to 30 micron size and the desiccant may be silica gel.

3. Demineralized water with a maximum inlet temperature of +65°C shall be used with, (1) specific resistivity maintained greater than 0.5 meg-cm, (2) oxygen content maintained less than 0.5 ppm, (3) an organic filter, and (4) a 3 to 4 micron filter at ambient tempeatures in excess of +4°C. A 60 percent ethylene glycol and 40 percent distilled water solution, by volume, with a maximum inlet temperature of +14°C, with the same resistivity, oxygen content, and filters shall be used at ambient temperatures of +4°C and lower.

4. The cathode terminal and high voltage bushing shall be insulated by immersion in oil that have the electrical characteristics specified in VV-I-530.

5. Extreme caution shall be used in the storage, handling, and installation of the tube to avoid imparting vibration or shock in excess of the values specified herein. To prevent demagnetization, ferromagnetic materials or energized magnets shall not be exposed to the tube magnet at a distance less than 12 inches.

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NOTES: -Continued

6. Government owned equipment installed at the contractors manufacturing facility shall be made available to perform all required tests specified herein. A transmitter test fixture with necessary microwave and modulator components, as specified in note 13, shall be used for these tests. Appropriate terminations shall be used where the TUT is required to operate into a 1.3:1 VSWR.

7. The TUT shall be initially set to operate at F3 at the average current (Ib) specified. The test frequency shall be manually changed without readjustment of high voltage.

8. The pulse characteristics determined in accordance with method 4304, shall be as follows:

trv: 1.5 µs (max) at 20 to 85 percent.
tfv: 5.0 µs (max) at 20 to 85 percent.

9. The following relationships shall exist between the TUT rf input and the rf output pulse:

(a) t1 is the time interval between the leading edge of the rf input pulse and the leading edge of the TUT rf output which shall be between zero (minimum) and 1.0 µs (maximum):
(b) The TUT rf envelopes shall be positioned to be totally contained within the rf drive (pd) pulse.
(c) The minimum isolation specified below shall be provided between the TUT and rf input source:

Frequency (MHz)	Isolation (dB)
2,600 to 3,200	20
3,200 to 3,400	12
3,400 to 3,500	8

10. The average rf drive (pd) to the TUT shall be adjusted to a value corresponding to 48 kw (peak) at the frequency of lowest drive power. The operating frequency shall be manually changed to each test frequency without readjustment of rf drive power.

11. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

12. The TUT shall be capable of satisfying all requirements specified herein at any frequency over the range F1 thru F3, even though specific frequencies are specified for these tests.

13. The tests shall be performed in a test circuit as shown on figure 3, with the TUT installed in a pulse transformer with a cavity damping sleeve and socket configuration as shown on figure 4, to overcome the effects of rf generated in the cathode well. The network impedance shall be within 5 percent of the specified value. The pulse width shall be the value specified minus zero +1 µs. (See note 6.)

14. The TUT shall be capable of rated power output with 48 kw peak drive power at any frequency in the range F1 through F3.

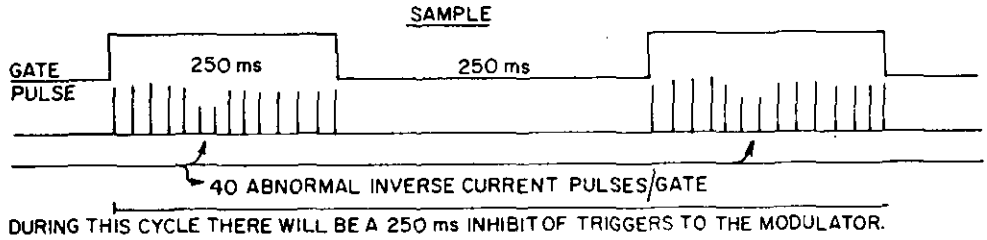
15. The operating frequency shall be swept from F1 through F3 at a rate of 100 sec/sweep. The duration of this test shall not exceed 5 minutes. The TUT shall be rejected for:

(a) Repeated waveguide breakdown, or
(b) Two successive inhibits of the modulator.

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NOTES: -Continued

16. Stability shall be measured in terms of the average number of output pulses missing, normally expressed as a percentage of the number of input pulses applied during the period of observation. Output(rf)pulses are considered to be "missing" if, due to any causes, the average rf pulse energy, within a ±1 percent frequency range of the normal operating frequency, is less than 70 percent of the energy of a normal pulse. The number of missing pulses shall not exceed the specified limit, when measured in any two alternating 250 millisecond time periods. Observation shall be made at the load phase position corresponding to maximum instability.



17. The mode stability test shall be conducted by adjusting the average current to 110 mAdc at F3, and manually sweeping the frequency from F3 to F1. The average current is then adjusted to 110 mAdc at F1, and the frequency manually swept from F1 to F3. The TUT shall be rejected if a mode jump occurs within this frequency range.

18. Service-life guarantee may be acceptable if specifically authorized by contract.

19. The phase of the mismatched load shall be adjusted for maximum reverse directed power at the input of the TUT.

20. The anode current shall be snapped on at the level required for the rated power output specified herein (see method 4250). The TUT shall be off for a minimum of 3 minutes prior to performing this test.

21. Life tests shall be conducted using the driver TUT in cascade with a final amplifier tube (8899). The applied voltage pulses shall be programmed such that the final amplifier is pulsed only on alternate pulses of the driver tube.

22. A life-test cycle shall consist of the following schedule:

Condition	Duration
High voltage "ON"	7.5 hours
High voltage "OFF"	0.5 hour

The frequency shall be varied from F1 through F3 and repeated after each 100 hours operating time.

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NOTES: -Continued

23. The TUT shall be mounted with the cathode axis vertical and vibrated as follows:

Exploratory:	Frequency	Amplitude (peak to peak)
	5 to 15 Hz	0.075 ±0.012 (0.150 total displacement)
	16 to 25 Hz	0.050 ±0.008 (0.100 total displacement)
	25 to 33 Hz	0.025 ±0.004 (0.050 total displacement)

Variable: As specified in the exploratory test except that the dwell time at each frequency with observed resonances shall be 5 minutes (approximately).

Endurance: As specified in the variable test except that the dwell time shall be 2 hours (minimum).

24. After the completion of this test, the TUT shall exhibit no mechanical failures and shall satisfy all requirements of the power output and stability tests of quality conformance inspection, part 1.

25. The TUT shall be mounted with the cathode axis vertical and shocked in accordance with condition A in a direction along the cathode axis. The TUT shall then be shocked in accordance with condition J (except 25 G) along the other two mutually perpendicular axes.

26. The TUT shall be placed in the test chamber in a manner similar to that in which it will be used in service, and such electrical and mechanical connections and instrumentations as may be necessary shall be completed. When changing chamber conditions from those required for one step to those required for any other step, the rates of temperature and pressure changes shall be as high as chamber construction permits, except that these rates shall not exceed 1°C (1.8°F) per second and 0.5 inch of mercury per second respectively. The change from room temperature to -62°C (-80°F) shall not exceed 4 hours. The temperature altitude conditions shall be as specified in the following steps:

Step	Chamber temperature a/	Pressure/altitude (simulated)	Operating time b/ (minutes)
1	-65°C (-85°F) (1 hr minimum)	Atmospheric	---
2	-54°C (-65°F)	Atmospheric	60
3	-54°C (-65°F)	30,000 ft	60
4	+85°C (185°F)	Atmospheric	120
5	+30°C (86°F)	Atmospheric	60
6	+55°C (130°F)	Atmospheric	4 cycles c/
7	+30°C (86°F)	Atmospheric	60
8	+66°C (151°F)	Atmospheric	15
9	+30°C (86°F)	30,000 ft	120
10	+59°C (138°F)	30,000 ft	30

a/ Thermally stabilize the TUT under nonoperating conditions.
b/ Monitor the power output with the TUT operating under conditions of test 1.
c/ Each cycle shall consist of 30 minutes "on" (operating) and 15 minutes "off".

27. In addition to the MIL-E-1 symbols, the following shall apply:

EG & W- - - - - - - - - - - Ethylene glycol and water
gpm - - - - - - - - - - - - Gallons per minute
P - - - - - - - - - - - - - Pressure
ΔP- - - - - - - - - - - - - Pressure drop
Pr- - - - - - - - - - - - - Reverse power

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Custodian:
Navy - EC

Review activity:
DSA - ES

User activities:
Navy - AS, CG, MC, OS, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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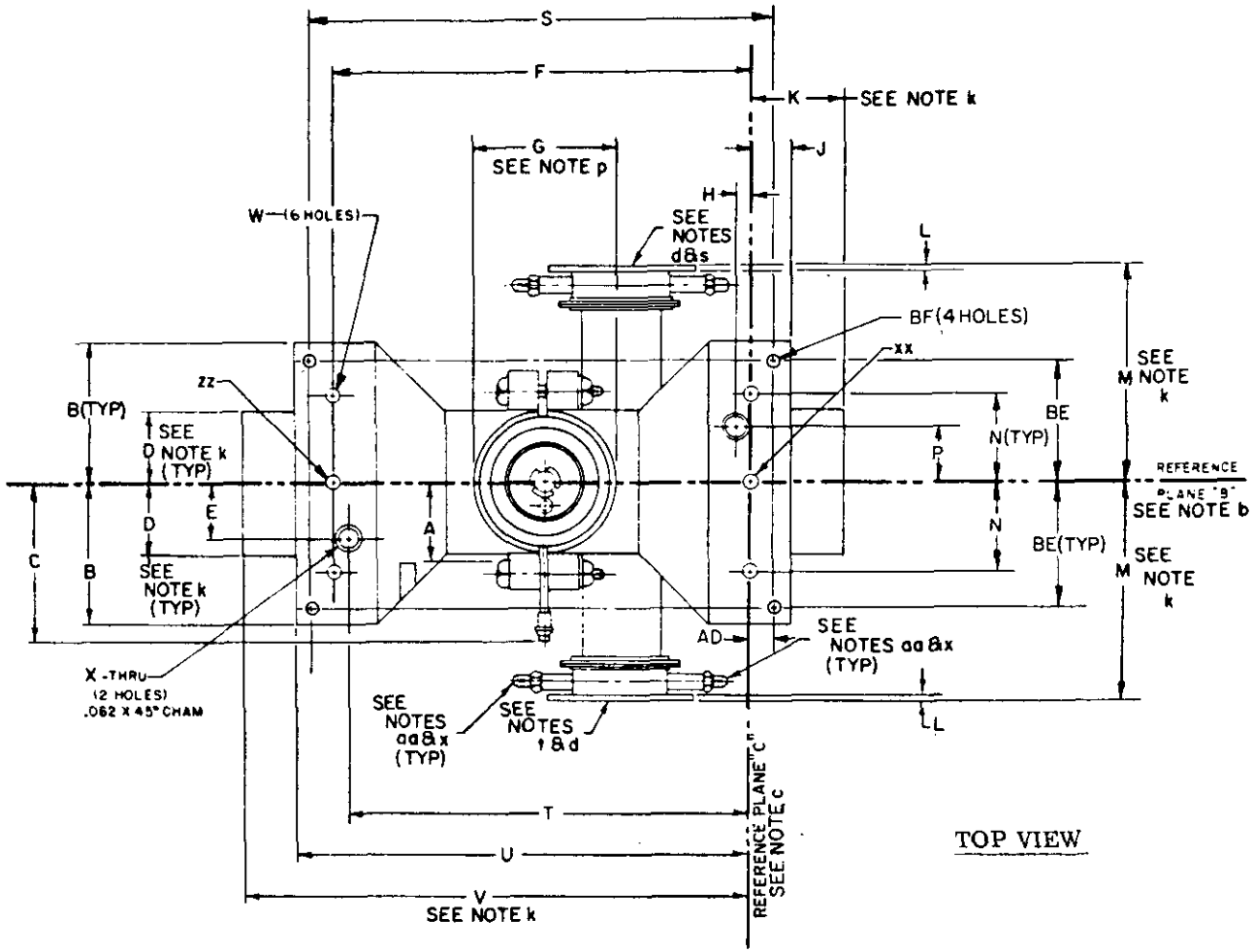


FIGURE 1. Outline drawing of electron tube type 8898.

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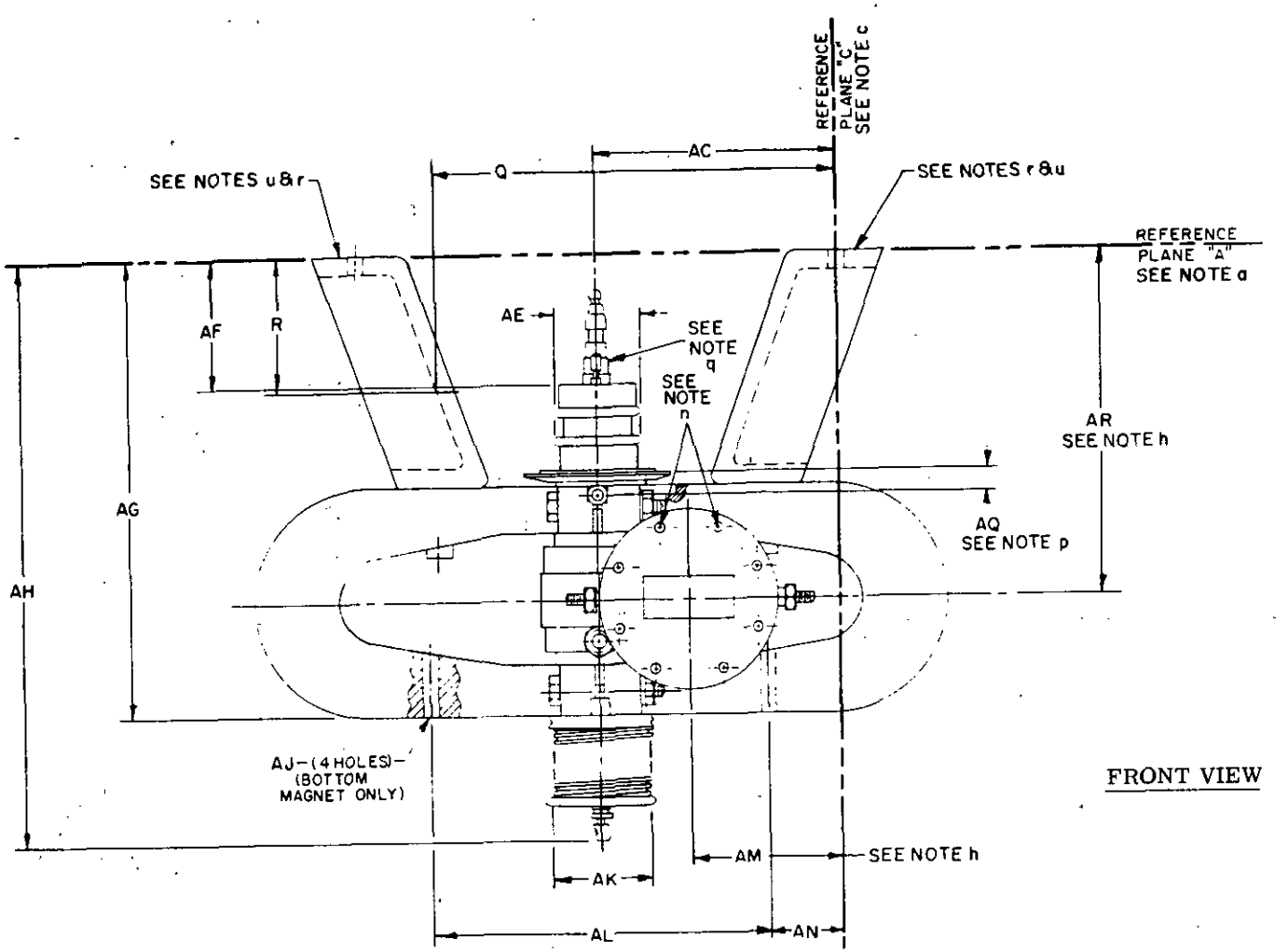


FIGURE 1. Outline drawing of electron tube type 8898 - Continued.

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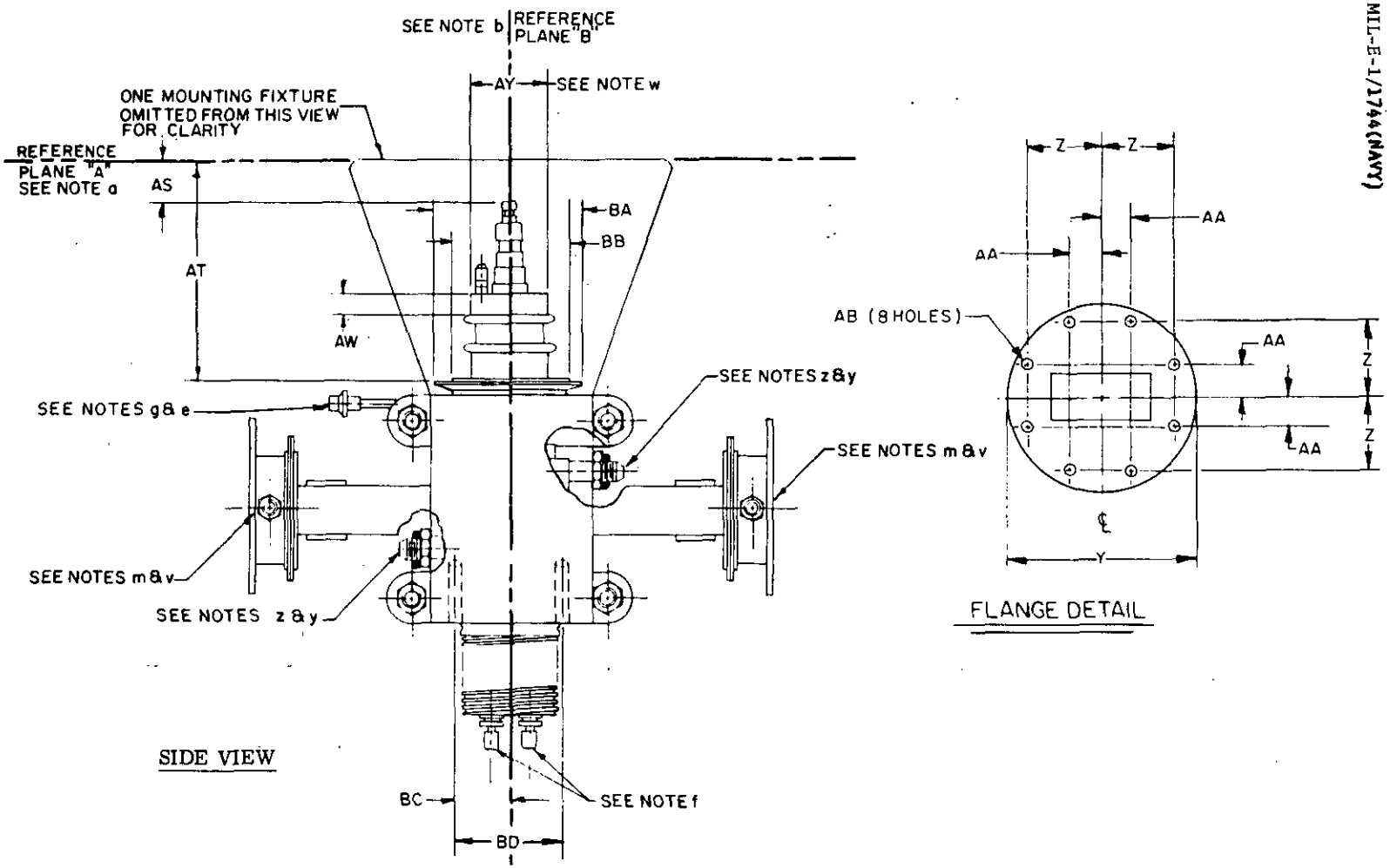


FIGURE 1. Outline drawing of electron tube type 8898 - Continued.

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L	DIMENSIONS			
T	INCHES		MILLIMETERS	
R	MIN	MAX	MIN	MAX
Quality conformance inspection, part 1 (SEE NOTE II)				
D		2.750		69.85
E	1.990	2.010	50.55	51.05
F	14.845	14.905	377.06	378.59
H	.552	.572	14.02	14.53
K		2.687		68.25
M	8.360	8.460	212.34	214.88
N	3.115	3.135	79.12	79.63
P	1.990	2.010	50.55	51.05
S	15.845	15.905	402.46	403.99
T	14.302	14.322	363.27	363.78
U		15.937		404.80
V		17.500		444.50
W	.576	.585	14.63	14.86
X	.593	.602	15.06	15.29
AB	.275	.291	6.99	7.39
AC	7.407	7.467	188.14	189.66
AD	.490	.510	12.45	12.95
AF	3.500	3.870	88.90	98.30
AG		14.500		368.30
AH		18.600		472.44
AJ	.425	.525	10.80	13.34
AL	10.410	10.590	264.41	268.99
AM	4.581	4.661	116.36	118.39
AQ	.435		11.05	
AR	10.430	10.490	264.92	266.45
AS	.812		20.62	
AT	6.400	6.650	162.56	168.91
BB		3.500		88.90
BD	2.910	3.090	73.91	78.49
BE	4.115	4.135	104.52	105.03
BF	.452	.459	11.48	11.66
Quality conformance inspection, part 2				
A		2.750		69.85
B		5.562		141.27
C		6.312		160.32
G	4.812		122.22	
J		1.062		26.97
L	.275	.325	6.99	8.26
Q	12.500		317.50	
R	4.052		102.92	
Y	5.300	5.325	134.62	135.28
Z	2.183	2.205	55.45	56.01
AA	.898	.920	22.81	23.37
AE		2.875		73.03
AK		3.250		82.55
AN	2.177	2.197	55.30	55.80
AW	.780		19.81	
AY	2.348	2.372	59.64	60.25
BA	4.485	4.515	113.92	114.68
BC	1.375	1.625	34.93	41.28

FIGURE 1. Outline drawing of electron tube type 8898 - Continued.

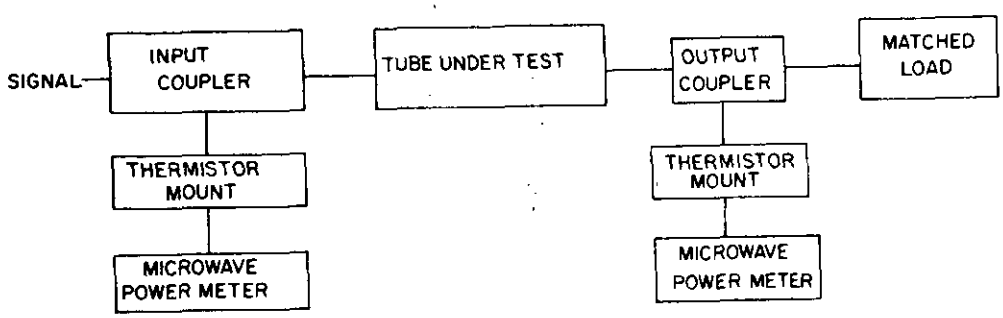
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NOTES:

a. Reference plane "A" is defined as a plane passing along the face of the mounting surface as shown.
b. Reference plane "B" is defined as a plane perpendicular to reference plane "A" passing through the ℄ of holes XX and ZZ at reference plane "A" as shown.
c. Reference plane "C" is defined as a plane mutually perpendicular to reference planes "A" and "B" passing through the ℄ of "XX" at reference plane "A" as shown.
d. This flange shall mate with UG-54B/U in accordance with MIL-F-3922/26.
e. The axis of this feature shall be parallel with or inclined away from reference plane "A".
f. Getter connections UG-106/U in accordance with REB 49236: Getter connections (red cap) and ground (black cap).
g. Wiggins BP1700 series nipple, or equivalent.
h. These dimensions apply to the ℄ of diameter "Y".
k. These dimensions include angular as well as lateral deviation.
m. These surfaces shall be flat within .015 in. (.38 mm).
n. These holes shall lie on a plane parallel to reference plane "A" within .025 in. (.64 mm).
p. Applies to uncoated area of magnet.
q. Cathode connection.
r. These surfaces shall be coplanar within .020 in. (.51 mm).
s. rf output.
t. rf input.
u. Mounting fixtures.
v. These surfaces shall be parallel with reference plane "B" within .050 in.(1.27 mm).
w. The axis of this diameter shall lie on reference plane "B" within .030 in.(.76 mm).
x. These couplings shall be AN815-4, or equivalent (stainless steel).
y. These couplings shall be AN815-8, or equivalent (stainless steel).
z. "O" ring shall be in accordance with MS28778-8 per MIL-G-5510.
aa. "O" ring shall be in accordance with MS28778-4 per MIL-G-5510.

FIGURE 1. Outline drawing of electron tube type 8898 - Continued.

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EQUIPMENT REQUIRED

1 - Cross-guide coupler (30-40 dB directivity)
1 - Microwave power meter
1 - Thermistor mount, 200 ohms negative
1 - Matched load
1 - Termination

NOTE: Use transmitter test fixture (notes 6 and 13).

FIGURE 2. Insertion loss test circuit.

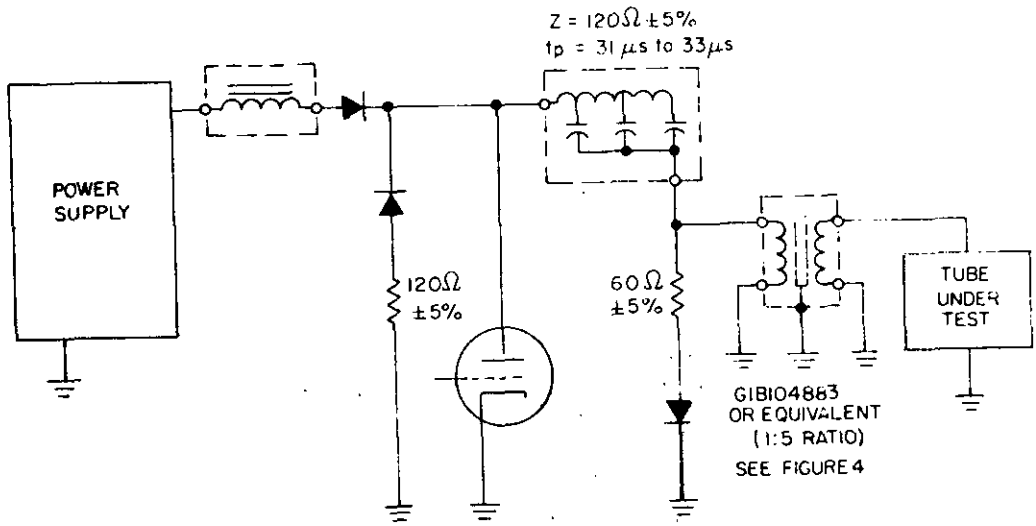
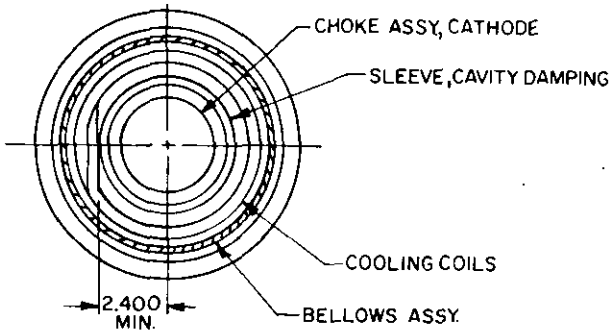
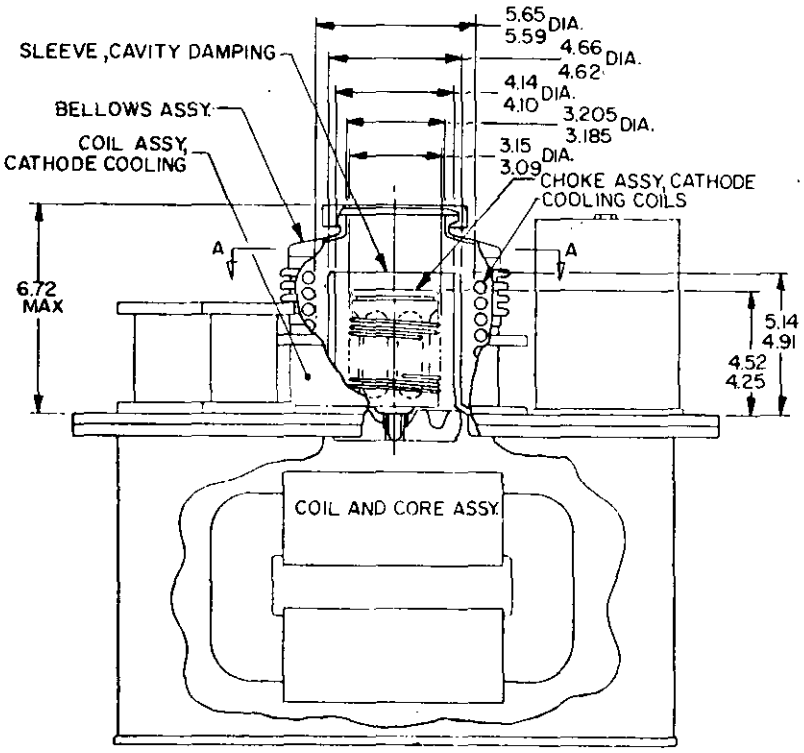


FIGURE 3. Electrical test circuit.

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SECTION A-A



INCHES	MM	INCHES	MM
2.400	60.96	4.52	114.81
3.09	78.49	4.62	117.35
3.15	80.01	4.66	118.36
3.185	80.90	4.91	124.71
3.205	81.41	5.14	130.56
4.10	104.14	5.59	141.99
4.14	105.16	5.65	143.51
4.25	107.95	6.72	170.69

FIGURE 4. Pulse transformer socket.

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NOTES:

The following describes the general characteristics of the rf pulse transformer to be used in conjunction with the testing of the TUT.

ELECTRICAL CHARACTERISTICS:

Turns ratio	1:5
Primary voltage (peak)	11 kV
Current (peak)	310A
Secondary voltage (peak)	55 kV
Current (peak)	62A

PULSE CHARACTERISTICS:

Pulse width (90 percent points)	27.5 μs
Rise time (10 percent to 90 percent)	1.5 μs max
Fall time (90 percent to 10 percent)	3.5 μs max
Overshoot and ripple	3 percent max
Droop	5 percent max

INSULATION:

Oil	In accordance with VV-I-530

MECHANICAL:

Sleeve, cavity damping	See figure Height 4.00 in. O.D. 4.66 in. I.D. 4.10 in. Composition: 23 PBW EPON 828 Resin, Shell Chemical Corp., or equal. 5 PBW Methane Diamine Catalyst. Shell Chemical Corp., or equal. 72 PBW No. 325 Mesh Powdered Iron. Van Waters and Rogers, Inc. No. 2226, or equal.
Choke assembly, cathode	28 turns #16 AWG wire. MIL-W-16878, type EE, @ .100 pitch on 3 inch dia. coil form. Coil form material: Epoxy resin laminated tubing, Nema Grade G-10, or equal.
Coil assembly, cathode cooling	7 turns of 3/8 O.D. seamless tubing. Cres 304, or equal.

FIGURE 4. Pulse transformer socket - Continued.

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